

SUBJECT 1 – QUANTUM MACHINE LEARNING

MODULE 1: Introduction to Quantum Machine Learning

Introduction to Machine Learning - Introduction to Quantum Computing - Introduction to Quantum Machine Learning - Classical Machine Learning vs Quantum Machine Learning - Quantum Advantage - Applications of Quantum Machine Learning - Challenges in Quantum Machine Learning

MODULE 2: Quantum Computing Foundations for Machine Learning

Qubits - Quantum States - Superposition - Entanglement - Quantum Measurement - Quantum Gates - Quantum Circuits - Multi-Qubit Systems - Quantum Information Encoding

MODULE 3: Quantum Data Encoding and Feature Maps

Classical Data Encoding - Quantum Data Encoding - Basis Encoding - Amplitude Encoding - Angle Encoding - Feature Maps - Quantum Feature Space - Quantum Kernels - Data Preparation for Quantum Machine Learning

MODULE 4: Variational Quantum Machine Learning

Parameterized Quantum Circuits - Variational Quantum Circuits - Trainable Parameters - Quantum Ansatz - Cost Functions - Classical Optimizers - Hybrid Quantum-Classical Learning - Training Variational Quantum Models

MODULE 5: Quantum Machine Learning Algorithms

Quantum Support Vector Machines - Quantum Kernel Methods - Quantum Neural Networks - Variational Quantum Classifiers - Quantum Clustering - Quantum Principal Component Analysis - Quantum Generative Models

MODULE 6: Quantum Neural Networks and Hybrid Models

Quantum Neural Networks - Quantum Layers - Quantum Perceptron's - Variational Quantum Neural Networks - Hybrid Quantum-Classical Models - Classical Neural Networks with Quantum Circuits - Training Hybrid Models - Loss Functions and Optimization

MODULE 7: Advanced Concepts in Quantum Machine Learning

Quantum Generative Learning - Quantum Reinforcement Learning - Quantum Transfer Learning - Quantum Ensemble Methods - Noise in Quantum Machine Learning - Quantum Error Mitigation - Scalability - Limitations and Future Directions